

Nicholas Barone

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WORK EXPERIENCE

Teaching Associate, Georgetown University August 2024 - May 2025

Taught one section of Calculus I per semester with an average class size of 28 students. Responsibilities including lesson planning, lecturing, quizzes, exams, and holding office hours. Outside of the classroom, responsibilities included hiring a teaching assistant and managing their hours and workload.

Teaching Assistant, Georgetown University August 2023 - May 2024

Worked as a teaching assistant for graduate Numerical Methods course, responsible for grading and helping students with homework questions.

Business Intelligence Analyst, Allen Lund Company July 2022 - July 2023

Created and built a truck freight market index tracking load frequency and cost variance dashboard for brokering to be distributed to managers of 37 offices, with the dashboard offering detailed operational information when focusing on specific shipping lanes and regions. Additionally created a proof of concept for an internal forecasting tool for predicting freight pricing using a few hundred parameters from imported and in-house historical data, incorporating based on seasonality using a sinusoidal model.

Global Financial Planning and Advising Intern, Citadel, LLC. May 2020 - December 2020

Built Alteryx applications to clean, organize, and perform calculations on firm-wide data from PowerBI SASS databases using DAX queries on monthly spending dashboards for COO's and department/fund managers to assist in operations and execution services. Provided data tie-outs with comments on variances for various COOs along with managing and fixing automated invoice systems for firm-wide telecommunications infrastructure in tens of millions of dollars monthly.

PROJECTS

2 Species in 2 Environment Competition and Stability

Under Supervision of Dr. Judith Miller at Georgetown University, the project aimed to model the competition and stability of 2 species that compete for resources in 2 distinct environment types accounting for births, deaths, and dispersal on discrete time intervals. Model, in MATLAB, simulates cases when only one species exists in a single patch, artificially introducing a species into a stable population in each patch, and starting with two random populations for each patch.

EDUCATION

2023 - 2025	PhD Student in Applied Mathematics; Georgetown University	
2023	MS Mathematics; Loyola University Chicago	
2022	BS Mathematics, Computer Science; Loyola University Chicago	Magna cum Laude

SKILLS

Mathematic Coursework	Dynamic Systems, Optimization, Differential Equations, Probability, Computational Mathematics, Cryptography, Deep Learning
Languages: Coding Applications	MATLAB, Python (TensorFlows/Keras), Java, Git, DAX PowerBI, Tableau, QlikSense, Alteryx